

Technical Report

DBST 651

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1. INTRODUCTION

The 48th Maintenance Group based out of RAF Lakenheath in UK is an aircraft maintenance unit supporting the United States Air Force European command. They maintain fighters and attack aircraft such as the F-15 Eagles and the newly acquired F-35 Lightning. Technicians from the 48th Maintenance Group collect and process data consisting of flight data information and maintenance transactions used in data analysis and key performance indicators calculations.

The F-35 weapon system is the most lethal, survivable and connected fighter aircraft in the Air Force inventory. The weapon system comes equipped with state-of-the-art systems including the Autonomic Logistics Information System (ALIS) which is utilized to managed missions, maintenance training, supply, and logistics data (Ninal, 2022). However, in a report by the Government Accountability Office (2020), ALIS was deemed inadequate and does not maintain accurate data. For this reason, a proposed database is introduced.

2. OVERVIEW

This technical report details the development of stand-alone database that will be utilize in tracking, recording, and managing information pertaining to the F-355 aircraft currently operated by the United States Air Forces in Europe. The THOR Database will serve as the primary management information system for the aircraft maintenance units in RAF Lakenheath, UK and future organizations operating the F-35 Lightnings. This document entails the database design decisions and its technical description of database code to help users implement and use the system. The goal of this report is to provide users and managers with information and details about the system in order to operate and maintain it efficiently.

3. LITERATURE REVIEW

A new system is currently in the works that will provide the missing capabilities that were promised by ALIS. The Operational Data Integrated Network or ODIN was a follow-on upgrade to ALIS introduced by Lockheed Martin (Osborn, 2022). The hardware was poised to be 75% smaller and lighter than the previous system and boasts a feature that utilizes sensors to examine the health of the aircraft and provides recommendation for solutions. This aims to anticipate component failures and take action before it could become a critical malfunction. According to Lockheed, ODIN is slated to be operational by February of 2022. However, the system has met some delays and we have yet to see a minimum viable product.

To supplement the lack of capability of ALIS, local maintenance organizations had to improvise and develop archaic process and tools to track and maintain data and information pertaining to the weapon system, so that managers can use data to drive decisions. The most popular approach is using Microsoft Excel based products and Microsoft Access database. One problem we see with such approach is that it isolates the data from the rest of the organization. While there are methods, of uploading multiple excel documents online for consumption, managing such process becomes cumbersome and poised to create many errors by lack of version controls. A better system such as an online SQL database will alleviate many of these issues and ensure that data is preserved and made available to common users.

4. ASSUMPTIONS

One cause of concern that could be problematic for this database is when data is entered to the system (database). One main assumption that we are making is that the users of the database will input the information correctly following the proper syntax and data format. For instance, under the SQUADRON table, the naming convention is not standardized in different squadrons. One squadron could use an

alphanumeric format like “493FS” and some may use “493rd FS”. This non-standardized way of inputting data could lead to issues in the future. Thus, it is imperative that we find a way to make the input standard. Typically, this can be done in the user application side by providing Dropdown boxes for the users to simply select. Another way to address this is to automatically input the correct “squadron” by simply associating the user to the squadron he or she is assigned to. However, this is currently outside the scope of this project, but it is worth to discuss early so that this issue can be mitigated.

5. DESIGN DECISIONS

Key Factors Influencing Design

The design of the database was inspired by the Air Force standard management information system that is currently being utilized by majority of maintenance organization. The Integrated Maintenance Database System or IMDS is the MIS currently in-placed that is utilized for maintenance data documentation for combat aircraft weapon systems such as the F-16 and the F-15 aircraft. The database is of entity-relational model that influences how it is maintained and operated. Using the same approach in the THOR Database, we hope that the users will easily learn to use the system due to the familiarity and similarities with the IMDS database.

THOR Database utilizes the same entity-relationship approach consisting of six entities related to each other. The Aircraft entity holds information about the F-3 aircraft that are currently assigned to the base. The Squadron entity holds all the information regarding the squadrons or units on a particular Air Force base. The Pilot entity is the personnel accountability table which holds people data about the pilots. The Flight data table is where flight information is recorded. The Aircraft Status entity holds information about the aircraft and could determine its flyable or non-flyable status. And lastly, the Maintenance entity is where maintenance work-orders are recorded.

The database also includes three different *views* to provide quick access queries to commonly searched information. The AircraftConfig view queries the aircraft and its current configuration – which tells the user what type of mission each aircraft can perform. The PilotRank view searches the database of pilots of certain ranks. And the LandingStatus view searches the database of aircraft and its landing status after its last flight.

Functional Design Decision

On typical operation, the database will be housed in a secured network approved for the level of information that it holds. The database must remain in an *unclassified secured network* to democratize the data available to its core users. Database access will be controlled via two-point authentication including Common Access Card (CAC) to access the Air Force's secure network and a User-ID/Password to access the frontend system via a web application. The web application will serve as the conduit that will be the entry point and access point of the DBMS. The database will also feed streaming or batch data to a dashboard system that provides data visualization key performance indicators to be used by managers for decision making.

Database Management System Decisions

The initial database was designed using Oracle SQL Developer version 21.2.1.204. Oracle SQL Developer is a proven database with outstanding track record, which is why it was selected for this project. It is also a common language that is employed in our organization so users will be able to adjust and use the database quickly and more effectively. Future state of the database will allow for additional entities or alterations using DDL and DML scripts to facilitate improvement and maintenance of the database. It is imperative that structural changes to the database is reflected in the entity-relational model to ensure the integrity of the database is preserved.

Security and Privacy Design Decisions

As mentioned earlier, the database will employ a two-factor authentication process using Common Access Card (CAC) to access the network and a user-id/Password login to access the database web application. Security and privacy of information must be protected to adhere to Department of Defense standards. The database is designed to maintain and preserve only those records that are allowed to be stored in the level of security the DOD system requires – IL5 level for information up to Confidential Unclassified Information (CUI).

6. STATEMENT OF WORK (SOW)

Executive Summary

The dedicated maintenance information system (MIS) designed for the F-35 Lightning aircraft presents limitation in democratizing data to analysts and maintenance managers. Currently, data collection efforts are done manually using Excel based products due to connectivity and data export limitation of the dedicated F-35 system. A separate database will be developed for the 48th Maintenance Group as a stopgap and fulfill the data processing needs of analysts and maintenance managers. The new database will reduce the manual data processing costs and increase productivity within the maintenance complex. Furthermore, it will streamline the ingestion, transformation, and serving of data to various business intelligence reports and dashboards enabling data-driven decisions in the organization.

Objective of the database project

The database will be used to track aircraft flight data and maintenance transactions for the maintenance unit. It will record the identity of the aircraft, flight take-off and landing times, maintenance status, location, and job workflow descriptions. It will also interact with an application server to collect, store, and retrieve aircraft flight data information and aircraft status information generated by technicians

from the field. Finally, the database will be used as the primary source of information for external dashboard applications via application program interfaces enabling up to date information for decision makers.

Project Scope

The database system will be implemented in a virtual machine simulating the Department of Defense Cloud environment to ensure the security of the stored data and allow for connections to other systems in the future. Interactions with input and output systems such as the web application for data entry will be simulated and will not be included in this implementation.

In-Scope Work

- The creation of the database to store aircraft flight data and maintenance information
- Creation of DML scripts to simulate connectivity to other systems
- Entity-Relationship Model
- Product Documentation
- Example DML Scripts

Out-of-Scope Work

- Data entry system to interact with the database
- External dashboard applications that collect and utilize data from the database

Database Goals, Expectations, and Deliverables

The database goals and expectations include simulating the following transactions: 1) Receiving aircraft flight data during pilot debrief. 2) Receiving aircraft maintenance status data from flightline maintenance Production Officers. 3) Generating reports that can be used for data analysis. The database

deliverables include the following: database, database documentation, entity-relationship diagrams (ERD), and scripts to simulate data retrieval transactions.

Database Benefits

Developing the database will reap immediate benefits such as decreasing the manual workload that data analysts and maintenance managers must do for simple analyses. It will allow faster data retrieval and enable management decision much faster than current processes. The database will also resolve the current lack of formal data storage for aircraft histories of the current information system. Furthermore, the database implementation can be utilized as the new standard for F-35 data reporting across other maintenance groups that operate these aircraft.

Project Hardware and Software Tools

This section discusses the hardware and software requirements that will be utilized in the development of this project. Discussion of the system for actual implementation is beyond the scope of this project.

Hardware

- Virtual Machine with Intel® Xeon® Platinum 8171M CPU @ 2.60GHz 210 GHZ

Software

- Windows 11 Pro Version 21H2 OS build: 22000.493
- Oracle SQL Developer Version 21.2.1.204
- Oracle Database Version 19.1.0 // 12.2.0.3
- Data Modeler Version 21.2.0.183 Build 183.1957
- Microsoft Visual Studio Code Version 1.71.2

Diagram Tool

- ER Assistant Version 2.10
- Visio 2016

Office Productivity

- Microsoft Office 365 App for the Enterprise

SQL Usage and Style

The following usage and style guide that will be the standard for this database implementation was derived from Simon Holywell's SQL Style Guide and Google's *Data Manipulation Language best practices*. It includes but not limited to the following and may be further refined to conform with Department of Defense's standards.

SQL Statement Structure

Script Format

- Use consistent and descriptive identifiers and names
- Use white space and indentation to make code easier to read
- Store ISO 8601 compliant time and date information (YYYY-MM-DD HH:MM:SS.SSSSS).
- Use only standard SQL functions instead of vendor specific functions
- Avoid redundant SQL code such as unnecessary quoting or parenthesis or WHERE clauses that can otherwise be derived.
- Avoid CamelCase and instead use snake_case
- Avoid prefixes such as sp_ or tbl

Comment Usage

- Include comments in SQL code where necessary. Use the C Style opening `/*` and closing `*/` where possible, otherwise use comments with `--` and finish with a new line

Object Naming Conventions

- Ensure unique names are used and does not conflict with reserved keywords
- Minimize the length to maximum of 30 bytes
- Begin names with letters and not end with underscore
- Do not sure special characters in names
- Use underscore in place of space
- Abbreviations are allowed as it is commonly used in the organization but ensure they understood across the organization

DDL and DML Statements

- Use WHERE clause to reduce the scope of locks
- Don't use DML statements and mutations in the same transaction
- Use `PENDING_COMMIT_TIMESTAMP` function to wrote commit timestamps
- Partitioned DML and date timestamps functions
- Use batch DML to make multiple writes with a single request

Student-Defined Sections and Additional Content

This portion discusses additional information regarding the project development approach. The aim is to outline the best possible software development methodology to avoid scope creep and ensure customer satisfaction via consistent feedback.

Software Development Methodology

Implementation of the database system should use a software development methodology approach consistent with DoD standards. As stated in DoD Standard 5000.2 released in 2000 (UMGC Wiki, 2022) Waterfall type methodologies should be discouraged. Instead, an iterative and incremental approach is encouraged. For this project, aim to use an Agile methodology for the development of the database such as SCRUM for improved productivity and consistent feedback loop, or Lean Software Development methodology for a minimalist but functional approach to save on cost and resources.

7. REQUIREMENTS DEFINITION DOCUMENT

Business Rules

1. PILOT may be assigned to SQUADRON
2. SQUADRON may be assigned to one PILOT
3. PILOT may command one or many FLIGHTS
4. Zero or many FLIGHTS may be commanded by one PILOT
5. AIRCRAFT may have zero or many FLIGHTS
6. One FLIGHT may only have one AIRCRAFT
7. One AIRCRAFT may have zero or many MAINTENANCE
8. A MAINTENANCE may only have one AIRCRAFT
9. An AIRCRAFT must have or many AIRCRAFT_STATUS
10. Many AIRCRAFT_STATUS may be loaded for one AIRCRAFT
11. Zero or many AIRCRAFT may be assigned to one SQUADRON
12. One SQUADRON may be assigned to one AIRCRAFT

Entity and Attribute Description

Entities

Entity Name: AIRCRAFT

Entity Description: The Aircraft entity maintains pertinent information about the aircraft.

Main Attributes:

- Aircraft_ID: (Primary Key) This is the unique number assigned to an aircraft.
- Configuration: A three-character combination of different aircraft configurations
- GEOLOC: A five-digit character indicating the aircraft's current location
- Mission_Design: an aircraft designator to identify the aircraft's model.
- Tail_Number: Aircraft Tail_Number is the common name for the aircraft.

Entity Name: AIRCRAFT_STATUS

Entity Description: Describes the current and historical maintenance status of the aircraft.

Main Attributes:

- Aircraft_ID: (Primary Key) This is the unique number assigned to an aircraft.
- ETIC: Estimated date and time of when the aircraft will be returned to mission capable status.
- Status: Alpha characters indicating the aircraft's current and historical status (i.e., FMC, NMCM, NMCS, NMCB, PMC)
- Status_Remarks: Additional remarks pertaining to the current status
- Status_Start_Date_Julian: Julian date equivalent of the date when the aircraft status was assigned.

- Status_Start_Time: The time the status started
- Status_Stop_Date_Julian: Julian date equivalent of the date when the aircraft status was terminated.
- Status_Stop_Time: The time the status ended

Entity Name: FLIGHT_DATA

Entity Description: Flight_Data entity records the flight information that the aircraft performed.

Main Attributes:

- Flight_ID: (Primary Key) Unique identifier of the Flight_Data entity. It is a combination of the Flight_Number and Flight_Julian_Date
- Flight_Julian_Date: Julian date equivalent of date. It is in an integer form.
- Flight_Number: is the assigned number for a specific flight.
- Landing_ICAO: International airport codes of the where the aircraft landed
- Landing_Status: an indicator of depicting the state of the aircraft after completion of the flight.
The following codes are used: Code 1 (No Issue), Code 2 (Minor maintenance issue), Code 3 (Major maintenance issue), Code 4 (Aircraft in-flight abort)
- Landing_Time: The time the flight is scheduled to land.
- Pilot_ID: (Foreign Key) Identifies who flew the sortie. Linked to Pilot entity.
- TakeOff_ICAO: International airport codes of the where the aircraft took-off.
- TakeOff_Time: It is the time the flight is scheduled to take-off

Entity Name: MAINTENANCE

Entity Description: Records maintenance records for each aircraft

Main Attributes:

- Job_Number: (Primary Key) a nine numeric character identifying the maintenance work order. it is in a format of 5-digit Julian date and arbitrary 4 numbers
- Aircraft_ID: (Foreign Key) a nine numeric character identifying the maintenance work order. it is in a format of 5-digit Julian date and arbitrary 4 numbers
- Aircraft_System_Code: Mandatory alpha numeric code indicating the part or component of the aircraft needing repair (i.e., 24A00 for Aux Power Unit)
- Corrective_Action: Narrative describing the fix performed to correct the issue
- Discrepancy_Narrative: Narrative describing the maintenance issue
- Job_Start_Date: Start date of the maintenance work order. Can be null if awaiting technicians.
- Job_Stop_Date: Completion date of the job work order

Entity Name: PILOT

Entity Description: Pilot entity holds the information about the pilots that flew the airplane.

Main Attributes:

- Pilot_ID: (Primary Key) Pilot identification consisting of initials and 4 arbitrary numbers.
- Squadron_ID: (Foreign Key) Squadron that the pilot is assigned to.
- First_Name: The pilot's first name
- Last_Name: Last name of the pilot.
- Qualified: A Yes or NO identifier indicating whether the pilot is current on his or her training qualifications.

- Rank: The pilot's rank following generic USAF rank mnemonic convention (i.e., 1LT, 2LT, CPT, MAJ, LCL, COL, GEN)

Entity Name: SQUADRON

Entity Description: Squadron is an entity about the aircraft flying unit that the aircraft is assigned to.

Main Attributes:

- Squadron_ID: (Primary Key) an identifiable information for which the Aircraft is assigned to. It is linked to the Squadron table.
- Country: Country location of the squadron
- Major_Command: Three letter indicator of the major command the squadron is assigned to (i.e., ACC for Air Combat Command)
- Squadron_Name: Common name of the squadron (i.e., 493 Fighter Squadron)
- State: State location of the squadron.
- Wing: The Air Force base the squadron is assigned to (i.e., 48 Fighter Wing)

Relationship and Cardinality Description

Relationship: PILOT is assigned to SQUADRON

Cardinality: 1:M relationship between PILOT and SQUADRON

Business rule: A zero or many PILOTS may be assigned to SQUADRON: One SQUADRON may be assigned to one PILOT

Relationship: PILOT commanded FLIGHT

Cardinality: 1:M relationship between PILOT and FLIGHT_DATA

Business rule: A PILOT may command one or many FLIGHTS; zero or many FLIGHTS may be commanded by one PILOT

Relationship: AIRCRAFT flew FLIGHT_DATA

Cardinality: 1:M relationship between FLIGHT_DATA and AIRCRAFT

Business rule: one AIRCRAFT may have zero or many FLIGHT_DATA; a FLIGHT_DATA may only have one AIRCRAFT

Relationship: AIRCRAFT in repair MAINTENANCE

Cardinality: 1:M relationship between AIRCRAFT and MAINTENANCE

Business rule: one AIRCRAFT may have zero or many MAINTENANCE; a MAINTENANCE may only have one AIRCRAFT

Relationship: AIRCRAFT has an AIRCRAFT_STATUS

Cardinality: 1:M relationship between AIRCRAFT and AIRCRAFT_STATUS

Business rule: one AIRCRAFT must have one or many AIRCRAFT_STATUS; many AIRCRAFT_STATUS may be loaded for one AIRCRAFT

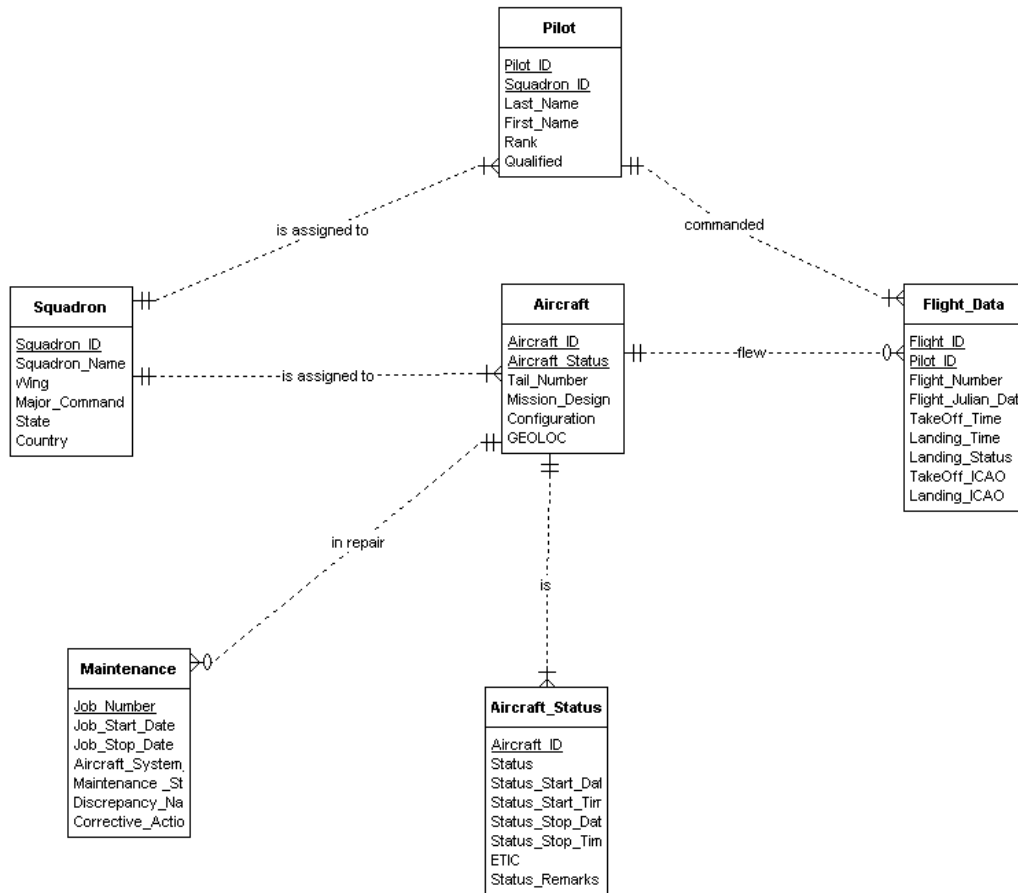
Relationship: AIRCRAFT is assigned to SQUADRON

Cardinality: 1:M relationship between AIRCRAFT and SQUADRON

Business rule: A zero or many AIRCRAFT may be assigned to one SQUADRON: One SQUADRON may be assigned to one AIRCRAFT

DETAILED DATABASE DESIGN

Entity Relationship Diagram (ERD)



DDL Source Code Embedded

/* DROP statements to clean up objects from previous run */

-- Tables

DROP TABLE Maintenance;

DROP TABLE Aircraft_Status;

DROP TABLE Flight_Data;

DROP TABLE Pilot;

DROP TABLE Aircraft;

DROP TABLE Squadron;

-- Indices

DROP INDEX idx_Tail;

DROP INDEX idx_Squadron;

DROP INDEX idx_Flight;

DROP INDEX idx_Status;

DROP INDEX idx_Squadron;

DROP INDEX idx_Pilot;

-- Views

DROP VIEW AircraftConfig;

DROP VIEW PilotRank;

DROP VIEW LandingStatus;

-- Sequences

DROP SEQUENCE seq_Aircraft_aircraft_id;

DROP SEQUENCE seq_Maintenance_job_id;

DROP SEQUENCE seq_Pilot_pilot_id;

-- Triggers

DROP TRIGGER trg_Flight;

DROP TRIGGER trg_Maintenance;

DROP TRIGGER trg_Status;

DROP TRIGGER trg_Aircraft;

DROP TRIGGER trg_Pilot;

/* Create tables */

CREATE TABLE Squadron (

 Squadron_ID INTEGER NOT NULL,

 Country VARCHAR2(30 CHAR),

 Major_Command VARCHAR2(50 CHAR) NOT NULL,

Squadron_Name VARCHAR2(50 CHAR) NOT NULL,
State VARCHAR2(20 CHAR),
Wing VARCHAR2(50 CHAR) NOT NULL,

CONSTRAINT PK_Squadron PRIMARY KEY (Squadron_ID));

CREATE TABLE Aircraft (

Aircraft_ID INTEGER NOT NULL,
Squadron_ID INTEGER NOT NULL,
Configuration CHAR (5 CHAR) NOT NULL,
GEOLOC VARCHAR2 (10 CHAR) NOT NULL,
Mission_Design VARCHAR2 (10 CHAR) NOT NULL,
Tail_Number VARCHAR2 (5 CHAR) NOT NULL,

CONSTRAINT PK_Aircraft PRIMARY KEY (Aircraft_ID),

CONSTRAINT FK_Aircraft_Squadron_ID FOREIGN KEY (Squadron_ID) References Squadron);

CREATE TABLE Pilot (

First_Name VARCHAR2(30 CHAR) NOT NULL,
Last_Name VARCHAR2(30 CHAR) NOT NULL,
Pilot_ID CHAR(7) NOT NULL,
Squadron_ID INTEGER NOT NULL,
Qualified VARCHAR2(1 CHAR),
Rank CHAR (3) NOT NULL,

CONSTRAINT PK_Pilot PRIMARY KEY (Pilot_ID),

CONSTRAINT FK_Pilot_Squadron_ID FOREIGN KEY (Squadron_ID) References Squadron,

CONSTRAINT ck_Pilot_Qualified CHECK (Qualified IN (1,0)));

CREATE TABLE Flight_Data (

Flight_ID INTEGER NOT NULL,

Aircraft_ID INTEGER NOT NULL,
Pilot_ID CHAR(7) NOT NULL,
Flight_Julian_Date INTEGER NOT NULL,
Flight_Number INTEGER NOT NULL,
Landing_ICAO VARCHAR2 (4 CHAR),
Landing_Status INTEGER NOT NULL,
Landing_Time VARCHAR2 (5 CHAR),
TakeOff_ICAO VARCHAR2 (4 CHAR),
TakeOff_Time VARCHAR2 (5 CHAR),

CONSTRAINT PK_Flight_Data PRIMARY KEY (Flight_ID),
CONSTRAINT FK_Flight_Data_Aircraft_ID FOREIGN KEY (Aircraft_ID) References Aircraft,
CONSTRAINT FK_Flight_Data_Pilot_ID FOREIGN KEY (Pilot_ID) References Pilot);

CREATE TABLE Aircraft_Status (

Aircraft_ID INTEGER NOT NULL,
ETIC VARCHAR2 (30 CHAR),
Status VARCHAR2 (5 CHAR) NOT NULL,
Status_Remarks VARCHAR2 (100 CHAR),
Status_Start_Date_Julian VARCHAR2 (6 CHAR) NOT NULL,
Status_Start_Time VARCHAR2 (30 CHAR) NOT NULL,
Status_Stop_Date_Julian VARCHAR2 (6 CHAR) NOT NULL,
Status_Stop_Time VARCHAR2 (30 CHAR) NOT NULL,

CONSTRAINT FK_Aircraft_Status_Aircraft_ID FOREIGN KEY (Aircraft_ID) References Aircraft);

CREATE TABLE Maintenance (

Aircraft_ID INTEGER NOT NULL,
Aircraft_System_Code VARCHAR2 (10 CHAR),
Corrective_Action VARCHAR2 (200 CHAR),

Discrepancy_Narrative VARCHAR2 (200 CHAR),

Job_Number VARCHAR2 (10 CHAR) NOT NULL,

Job_Start_Date VARCHAR2 (6 CHAR),

Job_Stop_Date VARCHAR2 (6 CHAR),

CONSTRAINT PK_Maintenance PRIMARY KEY (Job_Number),

CONSTRAINT FK_Maintenance_Aircraft_ID FOREIGN KEY (Aircraft_ID) References Aircraft);

/* Alter tables by adding audit columns */

```
ALTER TABLE Aircraft ADD (  
    created_by VARCHAR2(30),  
    date_created Date,  
    modified_by VARCHAR2(30),  
    date_modified DATE);
```

```
ALTER TABLE Pilot ADD (  
    created_by VARCHAR2(30),  
    date_created Date,  
    modified_by VARCHAR2(30),  
    date_modified DATE);
```

```
ALTER TABLE Squadron ADD (  
    created_by VARCHAR2(30),  
    date_created Date,  
    modified_by VARCHAR2(30),  
    date_modified DATE);
```

```
ALTER TABLE Maintenance ADD (  
    created_by VARCHAR2(30),
```

```
date_created Date,  
modified_by VARCHAR2(30),  
date_modified DATE);
```

```
ALTER TABLE Aircraft_Status ADD (  
    created_by VARCHAR2(30),  
    date_created Date,  
    modified_by VARCHAR2(30),  
    date_modified DATE);
```

```
ALTER TABLE Flight_Data ADD (  
    created_by VARCHAR2(30),  
    date_created Date,  
    modified_by VARCHAR2(30),  
    date_modified DATE);
```

```
/* Creating INDEX for commonly used columns and Natural Keys */
```

```
-- Natural Keys
```

```
CREATE INDEX idx_Tail  
ON Aircraft (Tail_Number);
```

```
CREATE INDEX idx_Wing  
ON Squadron (Wing);
```

```
CREATE INDEX idx_Flight  
ON Flight_Data (Flight_Number);
```

```
CREATE INDEX idx_Maintenance  
ON Maintenance (Aircraft_System_Code);
```

-- Foreign Keys

```
CREATE INDEX idx_Status  
ON Aircraft_Status (Aircraft_ID);
```

-- Frequently Queried Columns

```
CREATE INDEX idx_Pilot  
ON Pilot (Last_name, First_Name);
```

```
CREATE INDEX idx_Squadron  
ON Squadron (Squadron_Name);
```

/* Create two Views */

/* Create a view to query aircraft that has corresponding configuration code
taht will allow users to look for aircraft with proper configuration for a mission. */

```
CREATE OR REPLACE VIEW AircraftConfig AS  
SELECT Tail_Number, Mission_Design, GEOLOC  
FROM Aircraft  
WHERE Configuration = 'XX001';
```

/* Create a view to query pilots that are qualified to fly. This view allows users to query
pilots of certain qualification and provide easy future update as pilot changes in qualification. */

```
CREATE OR REPLACE VIEW PilotRank AS  
SELECT Last_Name, First_Name, Rank, Qualified  
FROM Pilot  
WHERE Qualified = '1';
```

/* Create a view for flights with certain Landing Status. This view will allow ease

of searching for flights that had maintenance issues upon landing. */

CREATE OR REPLACE VIEW LandingStatus AS

SELECT Flight_ID, Flight_Number, Landing_Status

FROM Flight_Data

WHERE Landing_Status = '3';

/* Create Sequences */

CREATE SEQUENCE seq_Aircraft_aircraft_id

INCREMENT BY 1

START WITH 001

NOMAXVALUE

MINVALUE 001

NOCACHE;

CREATE SEQUENCE seq_Maintenance_job_id

INCREMENT BY 1

START WITH 1000

NOMAXVALUE

MINVALUE 1000

NOCACHE;

CREATE SEQUENCE seq_Pilot_pilot_id

INCREMENT BY 1

START WITH 0001

NOMAXVALUE

MINVALUE 0001

NOCACHE;

/* Create triggers */

```
/* trg_Flight trigger automatically updates the created_by, date_created,  
modified_by, and date_modified fields with current values when INSERT or UPDATE occurs */
```

```
CREATE OR REPLACE TRIGGER trg_Flight  
  BEFORE INSERT OR UPDATE ON Flight_Data  
  FOR EACH ROW  
  BEGIN  
    IF INSERTING THEN  
      IF :NEW.created_by is NULL THEN  
        :NEW.created_by := USER;  
      END IF;  
      IF :NEW.date_created is NULL THEN  
        :NEW.date_created := SYSDATE;  
      END IF;  
    END IF;  
    IF INSERTING OR UPDATING THEN  
      :NEW.modified_by := USER;  
      :NEW.date_modified := SYSDATE;  
    END IF;  
  END;  
/
```

```
/* trg_Maintenance trigger automatically updates the created_by, date_created,  
modified_by, and date_modified fields with current values when INSERT or UPDATE happened */
```

```
CREATE OR REPLACE TRIGGER trg_Maintenance  
  BEFORE INSERT OR UPDATE ON Maintenance  
  FOR EACH ROW  
  BEGIN  
    IF INSERTING THEN  
      IF :NEW.created_by is NULL THEN  
        :NEW.created_by := USER;  
      END IF;  
    END IF;  
  END;
```

```
        END IF;

        IF :NEW.date_created is NULL THEN

            :NEW.date_created := SYSDATE;

        END IF;

    END IF;

    IF INSERTING OR UPDATING THEN

        :NEW.modified_by := USER;

        :NEW.date_modified := SYSDATE;

    END IF;

END;

/
```

/* trg_Status trigger automatically updates the created_by, date_created, modified_by, and date_modified fields with current values when INSERT or UPDATE happened */

```
CREATE OR REPLACE TRIGGER trg_Status

BEFORE INSERT OR UPDATE ON Aircraft_Status

FOR EACH ROW

BEGIN

    IF INSERTING THEN

        IF :NEW.created_by is NULL THEN

            :NEW.created_by := USER;

        END IF;

        IF :NEW.date_created is NULL THEN

            :NEW.date_created := SYSDATE;

        END IF;

    END IF;

    IF INSERTING OR UPDATING THEN

        :NEW.modified_by := USER;

        :NEW.date_modified := SYSDATE;

    END IF;

END;
```

END;

/

/* trg_Aircraft trigger automatically updates the created_by, date_created,
modified_by, and date_modified fields with current values when INSERT or UPDATE happened */

CREATE OR REPLACE TRIGGER trg_Aircraft

BEFORE INSERT OR UPDATE ON Aircraft

FOR EACH ROW

BEGIN

IF INSERTING THEN

IF :NEW.created_by is NULL THEN

:NEW.created_by := USER;

END IF;

IF :NEW.date_created is NULL THEN

:NEW.date_created := SYSDATE;

END IF;

END IF;

IF INSERTING OR UPDATING THEN

:NEW.modified_by := USER;

:NEW.date_modified := SYSDATE;

END IF;

END;

/

/* trg_Pilot trigger automatically updates the created_by, date_created,
modified_by, and date_modified fields with current values when INSERT or UPDATE happened */

CREATE OR REPLACE TRIGGER trg_Pilot

BEFORE INSERT OR UPDATE ON Pilot

FOR EACH ROW

BEGIN

```
IF INSERTING THEN
    IF :NEW.created_by is NULL THEN
        :NEW.created_by := USER;
    END IF;
    IF :NEW.date_created is NULL THEN
        :NEW.date_created := SYSDATE;
    END IF;
END IF;
IF INSERTING OR UPDATING THEN
    :NEW.modified_by := USER;
    :NEW.date_modified := SYSDATE;
END IF;
END;
/

/* trg_Squadron trigger automatically updates the created_by, date_created,
modified_by, and date_modified fields with current values when INSERT or UPDATE happened */
CREATE OR REPLACE TRIGGER trg_Squadron
BEFORE INSERT OR UPDATE ON Squadron
FOR EACH ROW
BEGIN
    IF INSERTING THEN
        IF :NEW.created_by is NULL THEN
            :NEW.created_by := USER;
        END IF;
        IF :NEW.date_created is NULL THEN
            :NEW.date_created := SYSDATE;
        END IF;
    END IF;
    IF INSERTING OR UPDATING THEN
```

```
:NEW.modified_by := USER;

:NEW.date_modified := SYSDATE;

END IF;

END;

/

/* Demonstrate the successful creation of all aforementioned objects by
querying the database catalog/data dictionary */

SELECT TABLE_NAME FROM USER_TABLES;
SELECT OBJECT_NAME, STATUS, CREATED, LAST_DDL_TIME FROM USER_OBJECTS;

COMMIT;
```

DML and Query Source Code

```
/* Poupopulate all tables */

-- Squadron

INSERT INTO SQUADRON (squadron_id, country, major_command, squadron_name, state, wing)
VALUES ('100495', 'UK', 'USAFE', '495th Fighter Squadron', 'N/A', '48th Fighter Wing');
INSERT INTO SQUADRON (squadron_id, country, major_command, squadron_name, state, wing)
VALUES ('100496', 'UK', 'USAFE', '496th Fighter Squadron', 'N/A', '48th Fighter Wing');
INSERT INTO SQUADRON (squadron_id, country, major_command, squadron_name, state, wing)
VALUES ('100497', 'UK', 'USAFE', '497th Fighter Squadron', 'N/A', '48th Fighter Wing');
INSERT INTO SQUADRON (squadron_id, country, major_command, squadron_name, state, wing)
VALUES ('100498', 'UK', 'USAFE', '498th Fighter Squadron', 'N/A', '48th Fighter Wing');
INSERT INTO SQUADRON (squadron_id, country, major_command, squadron_name, state, wing)
VALUES ('100001', 'USA', 'ACC', 'First Fighter Squadron', 'VA', 'First Fighter Wing');
INSERT INTO SQUADRON (squadron_id, country, major_command, squadron_name, state, wing)
```

```
VALUES ('100086', 'Germany', 'USAFE', '786th Airlift Squadron', 'N/A', '86th Airlift Wing');
INSERT INTO SQUADRON (squadron_id, country, major_command, squadron_name, state, wing)
VALUES ('100052', 'Germany', 'USAFE', '52nd Fighter Squadron', 'N/A', '52nd Fighter Wing');
INSERT INTO SQUADRON (squadron_id, country, major_command, squadron_name, state, wing)
VALUES ('100388', 'USA', 'ACC', '388th Fighter Squadron', 'Utah', '388th Fighter Wing');
INSERT INTO SQUADRON (squadron_id, country, major_command, squadron_name, state, wing)
VALUES ('100419', 'USA', 'ACC', '419th Fighter Squadron', 'N/A', '419th Fighter Wing');
INSERT INTO SQUADRON (squadron_id, country, major_command, squadron_name, state, wing)
VALUES ('100004', 'USA', 'ACC', '4th Fighter Squadron', 'North Carolina', '4th Fighter Wing');

-- Aircraft table
INSERT INTO AIRCRAFT (aircraft_id, squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034205589', '100495', 'XX001', 'MSET', 'F035A', 'A5589');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034205583', '100495', 'XX001', 'MSET', 'F035A', 'A5583');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034205581','100496', 'XX001', 'MSET', 'F035A', 'A5581');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034205571', '100496', 'XX001', 'MSET', 'F035A', 'A5571');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034205572', '100495', 'XX001', 'MSET', 'F035A', 'A5572');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034205570', '100496', 'XX001', 'MSET', 'F035A', 'A5570');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034195486', '100496', 'XX001', 'MSET', 'F035A', 'A5486');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034195485', '100495', 'XX001', 'MSET', 'F035A', 'A5485');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034195484', '100496', 'XX001', 'MSET', 'F035A', 'A5484');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
```

```
VALUES ('0034195475', '100497', 'XX001', 'MSET', 'F035A', 'A5475');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034195476', '100497', 'XX001', 'MSET', 'F035A', 'A5476');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034195474', '100497', 'XX001', 'MSET', 'F035A', 'A5474');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034195473', '100497', 'XX001', 'MSET', 'F035A', 'A5473');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034195495','100495', 'XX001', 'MSET', 'F035A', 'A5495');
INSERT INTO AIRCRAFT (aircraft_id,squadron_id, configuration, geoloc, mission_design, tail_number)
VALUES ('0034205580', '100496', 'XX001', 'MSET', 'F035A', 'A5580');
```

-- Pilot table

```
INSERT INTO PILOT (first_name, last_name, pilot_id, squadron_id, qualified, rank)
VALUES ('Richard', 'Galero', '1234567', '100495', '1', 'O03');
INSERT INTO PILOT (first_name, last_name, pilot_id, squadron_id, qualified, rank)
VALUES ('James', 'Rivas', '0000002','100496', '1', 'O02');
INSERT INTO PILOT (first_name, last_name, pilot_id, squadron_id, qualified, rank)
VALUES ('Kyle', 'Ward', '0000003', '100497', '1', 'O04');
INSERT INTO PILOT (first_name, last_name, pilot_id, squadron_id, qualified, rank)
VALUES ('Joseph', 'Miles', '0000004', '100495', '1', 'O05');
INSERT INTO PILOT (first_name, last_name, pilot_id, squadron_id, qualified, rank)
VALUES ('Joel', 'Banks', '0000005', '100496', '1', 'O05');
INSERT INTO PILOT (first_name, last_name, pilot_id, squadron_id, qualified, rank)
VALUES ('Benjamin', 'Lovett', '0000006', '100497', '1', 'O03');
INSERT INTO PILOT (first_name, last_name, pilot_id, squadron_id, qualified, rank)
VALUES ('Paul', 'Walker', '0000007', '100495', '0', 'O04');
INSERT INTO PILOT (first_name, last_name, pilot_id, squadron_id, qualified, rank)
VALUES ('Leila', 'Thompson', '0000008', '100496', '1', 'O03');
INSERT INTO PILOT (first_name, last_name, pilot_id, squadron_id, qualified, rank)
```



```
VALUES ('Jonathan', 'Dayao', '0000009', '100497', '1', 'O03');
```

```
INSERT INTO PILOT (first_name, last_name, pilot_id, squadron_id, qualified, rank)
```

```
VALUES ('Roseanne', 'Garcia', '0000010', '100495', '1', 'O05');
```

```
INSERT INTO PILOT (first_name, last_name, pilot_id, squadron_id, qualified, rank)
```

```
VALUES ('Kristin', 'Lopez', '0000011', '100496', '0', 'O03');
```

```
-- Flight Data
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339801', '0034195475', '0000002', '22339', '801', 'EGUL', '1', '14:32', 'EGUL', '13:00');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339802', '0034195486', '0000003', '22339', '802', 'EGUL', '1', '14:33', 'EGUL', '13:00');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339803', '0034205572', '0000004', '22339', '803', 'EGUL', '1', '14:32', 'EGUL', '13:00');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339804', '0034195474', '0000005', '22339', '804', 'EGUL', '1', '14:27', 'EGUL', '13:00');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339805', '0034195476', '0000006', '22339', '805', 'EGUL', '1', '14:50', 'EGUL', '13:00');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339806', '0034205581', '0000007', '22339', '806', 'EGUL', '1', '14:42', 'EGUL', '13:06');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339807', '0034195495', '0000008', '22339', '807', 'EGUL', '1', '14:41', 'EGUL', '13:04');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339808', '0034205571', '0000009', '22339', '808', 'EGUL', '1', '14:41', 'EGUL', '13:05');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339809', '0034205580', '0000010', '22339', '809', 'EGUL', '1', '14:42', 'EGUL', '13:08');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339810', '0034205589', '0000011', '22339', '810', 'EGUL', '1', '14:42', 'EGUL', '13:08');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339811', '0034195475', '0000002', '22339', '811', 'EGUL', '1', '17:20', 'EGUL', '19:10');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339812', '0034195486', '0000003', '22339', '812', 'EGUL', '1', '17:20', 'EGUL', '19:10');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339813', '0034195474', '0000004', '22339', '813', 'EGUL', '1', '17:20', 'EGUL', '19:10');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339814', '0034195476', '0000005', '22339', '814', 'TYFR', '1', '17:20', 'EGUL', '20:10');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339815', '0034205572', '0000006', '22339', '815', 'EGUL', '1', '17:34', 'EGUL', '19:23');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339816', '0034205581', '0000007', '22339', '816', 'EGUL', '1', '17:34', 'EGUL', '19:23');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339817', '0034205580', '0000008', '22339', '817', 'EGUL', '1', '17:30', 'EGUL', '19:17');
```

```
INSERT INTO FLIGHT_DATA(flight_id, aircraft_id, pilot_id, flight_julian_date, flight_number,  
landing_icao, landing_status, landing_time, takeoff_icao, takeoff_time)
```

```
VALUES ('22339818', '0034205571', '0000009', '22339', '818', 'EGUL', '1', '17:30', 'EGUL', '19:25');
```

-- Aircraft Status

```
INSERT INTO AIRCRAFT_STATUS (aircraft_id, ETIC, status, status_remarks, status_start_date_julian,  
status_start_time, status_stop_date_julian, status_stop_time)
```

```
VALUES ('0034205589', 'None', 'FMC', 'None', '22339', '01:00', '22339', 'None');
```

```
INSERT INTO AIRCRAFT_STATUS (aircraft_id, ETIC, status, status_remarks, status_start_date_julian,
status_start_time, status_stop_date_julian, status_stop_time)
VALUES ('0034205583', '+42 Hours', 'NMC', 'VMS Flight control inop', '22339', '07:00', '22340', '12:00');

INSERT INTO AIRCRAFT_STATUS (aircraft_id, ETIC, status, status_remarks, status_start_date_julian,
status_start_time, status_stop_date_julian, status_stop_time)
VALUES ('0034205581', 'None', 'FMC', 'None', '22339', '07:45', '22339', 'None');

INSERT INTO AIRCRAFT_STATUS (aircraft_id, ETIC, status, status_remarks, status_start_date_julian,
status_start_time, status_stop_date_julian, status_stop_time)
VALUES ('0034205581', 'None', 'FMC', 'None', '22339', '07:45', '22339', 'None');

INSERT INTO AIRCRAFT_STATUS (aircraft_id, ETIC, status, status_remarks, status_start_date_julian,
status_start_time, status_stop_date_julian, status_stop_time)
VALUES ('0034205571', 'None', 'FMC', 'None', '22339', '01:00', '22339', 'None');

INSERT INTO AIRCRAFT_STATUS (aircraft_id, ETIC, status, status_remarks, status_start_date_julian,
status_start_time, status_stop_date_julian, status_stop_time)
VALUES ('0034205572', 'None', 'FMC', 'None', '22339', '03:00', '22339', 'None');

INSERT INTO AIRCRAFT_STATUS (aircraft_id, ETIC, status, status_remarks, status_start_date_julian,
status_start_time, status_stop_date_julian, status_stop_time)
VALUES ('0034205570', 'Parts +4 hours', 'PMC', 'Landing gear requires inspection', '22339', '03:00',
'22339', 'None');

INSERT INTO AIRCRAFT_STATUS (aircraft_id, ETIC, status, status_remarks, status_start_date_julian,
status_start_time, status_stop_date_julian, status_stop_time)
VALUES ('0034195486', 'Parts + 24', 'NMC', 'landing gear tires require replace', '22339', '07:00', '22339',
'None');

INSERT INTO AIRCRAFT_STATUS (aircraft_id, ETIC, status, status_remarks, status_start_date_julian,
status_start_time, status_stop_date_julian, status_stop_time)
VALUES ('0034195485', 'None', 'FMC', 'None', '22339', '00:01', '22339', 'None');

INSERT INTO AIRCRAFT_STATUS (aircraft_id, ETIC, status, status_remarks, status_start_date_julian,
status_start_time, status_stop_date_julian, status_stop_time)
VALUES ('0034195484', 'None', 'FMC', 'None', '22339', '00:01', '22339', 'None');

INSERT INTO AIRCRAFT_STATUS (aircraft_id, ETIC, status, status_remarks, status_start_date_julian,
status_start_time, status_stop_date_julian, status_stop_time)
VALUES ('0034195475', 'None', 'FMC', 'None', '22339', '00:01', '22339', 'None');
```

-- Maintenance

```
INSERT INTO MAINTENANCE (aircraft_id, aircraft_system_code, corrective_action,
discrepancy_narrative, job_number, job_start_date, job_stop_date)
VALUES ('0034205583', '42000', 'In-Work', 'VMS Flight control inop', '223391201', '22339', 'None');

INSERT INTO MAINTENANCE (aircraft_id, aircraft_system_code, corrective_action,
discrepancy_narrative, job_number, job_start_date, job_stop_date)
VALUES ('0034205583', '42001', 'In-Work', 'aircraft anels require removal', '223391202', '22339', 'None');

INSERT INTO MAINTENANCE (aircraft_id, aircraft_system_code, corrective_action,
discrepancy_narrative, job_number, job_start_date, job_stop_date)
VALUES ('0034205583', '24000', 'R2 flight data recorder', 'Flight data recorder was intermittent',
'223391203', '22339', '22339');

INSERT INTO MAINTENANCE (aircraft_id, aircraft_system_code, corrective_action,
discrepancy_narrative, job_number, job_start_date, job_stop_date)
VALUES ('0034195486', '13000', 'In-Work', 'landing gear requires inspection', '223391204', '22339',
'None');

INSERT INTO MAINTENANCE (aircraft_id, aircraft_system_code, corrective_action,
discrepancy_narrative, job_number, job_start_date, job_stop_date)
VALUES ('0034195486', '13100', 'Removed tires for inspection', 'landing gear requires inspection',
'223391205', '22339', 'None');

INSERT INTO MAINTENANCE (aircraft_id, aircraft_system_code, corrective_action,
discrepancy_narrative, job_number, job_start_date, job_stop_date)
VALUES ('0034195486', '13110', 'Replaced main landing gear tires', 'landing gear requires inspection',
'223391206', '22339', '22339');

INSERT INTO MAINTENANCE (aircraft_id, aircraft_system_code, corrective_action,
discrepancy_narrative, job_number, job_start_date, job_stop_date)
VALUES ('0034205570', '04000', 'In-Work', '400 Hour inspection due', '223391207', '22339', 'None');

INSERT INTO MAINTENANCE (aircraft_id, aircraft_system_code, corrective_action,
discrepancy_narrative, job_number, job_start_date, job_stop_date)
VALUES ('0034205570', '04100', 'removed panels for 400 hr insp', '400 Hour inspection due',
'223391208', '22339', '22339');

INSERT INTO MAINTENANCE (aircraft_id, aircraft_system_code, corrective_action,
discrepancy_narrative, job_number, job_start_date, job_stop_date)
VALUES ('0034205570', '04500', 'In-Work', 'Inspect engine bay', '223391209', '22339', 'None');

INSERT INTO MAINTENANCE (aircraft_id, aircraft_system_code, corrective_action,
discrepancy_narrative, job_number, job_start_date, job_stop_date)
VALUES ('0034205570', '04800', 'In-Work', 'test hydraulic flaps', '223391210', '22339', 'None');
```

```
INSERT INTO MAINTENANCE (aircraft_id, aircraft_system_code, corrective_action,  
discrepancy_narrative, job_number, job_start_date, job_stop_date)  
VALUES ('0034205570', '04900', 'In-Work', 'aircraft wash', '223391211', '22339', 'None');
```

```
COMMIT;
```

```
-- 1. Select all columns and all rows from one table
```

```
SELECT * FROM aircraft;
```

```
-- 2. Select five columns and all rows from one table
```

```
SELECT rank, first_name, last_name, pilot_id, qualified  
FROM pilot;
```

```
-- 3. Select all columns from all rows from one view
```

```
SELECT * FROM PILOTRANK;
```

```
/* 4. Using a join on 2 tables, select all columns and all rows from the tables  
without the use of a Cartesian product */
```

```
SELECT * FROM flight_data  
INNER JOIN aircraft on aircraft.aircraft_id = flight_data.aircraft_id;
```

```
/* 5. Select and order data retrieved from one table */
```

```
SELECT * FROM squadron  
ORDER BY squadron_id ASC;
```

```
/* 6. Using a join on 3 tables, select 5 columns from the 3 tables. Use  
syntax that would limit the output to 10 rows */
```

```
SELECT aircraft.tail_number, aircraft_status.status,  
maintenance.aircraft_system_code, maintenance.discrepancy_narrative, maintenance.job_number  
FROM aircraft
```

```
INNER JOIN aircraft_status ON aircraft_status.aircraft_id = aircraft.aircraft_id  
INNER JOIN maintenance ON maintenance.aircraft_id = aircraft_status.aircraft_id;
```

```
/* 7. Select distinct rows using join on 3 tables */
```

```
SELECT DISTINCT aircraft.tail_number, aircraft.aircraft_id, status  
FROM aircraft_status  
INNER JOIN aircraft ON aircraft.aircraft_id = aircraft_status.aircraft_id  
INNER JOIN maintenance ON maintenance.aircraft_id = aircraft.aircraft_id;
```

```
/* 8. Use GROUP BY and HAVING in a select statement using one or more tables */
```

```
SELECT squadron_id, COUNT(tail_number) AS AircraftBySQ  
FROM aircraft  
GROUP BY squadron_id  
HAVING COUNT (tail_number) >= 3;
```

```
/* 9. Use IN clause o select data from one or more tables */
```

```
SELECT squadron_id, squadron_name, country  
FROM squadron  
WHERE country IN ('Germany', 'UK');
```

```
/* 10. Select length of one column from one table */
```

```
SELECT LENGTH(squadron_name)  
FROM squadron;
```

```
/* 11. Delete one record from a table with ROLLBACK */
```

```
SELECT *  
FROM flight_data  
WHERE flight_id = '22339801';
```

```
DELETE
```

```
FROM flight_data  
WHERE flight_id = '22339801';
```

```
SELECT *  
FROM flight_data  
WHERE flight_id = '22339801';
```

```
ROLLBACK;
```

```
SELECT *  
FROM flight_data  
WHERE flight_id = '22339801';
```

/* 12. Update one record from table. Use selects statements to demonstrate the table contents before and after the UPDATE statement */

```
SELECT *  
FROM squadron  
WHERE country = 'Germany';
```

```
UPDATE squadron  
SET major_command = 'ACC'  
WHERE squadron_id = '100052';
```

```
SELECT *  
FROM squadron  
WHERE country = 'Germany';
```

```
ROLLBACK;
```

```
SELECT *
```

FROM squadron

WHERE country = 'Germany';

/* 13. Perform 8 additional advanced queries */

/* 1. Select flights flown by non-qualified pilots JOIN two tables with WHERE clause

this query joins flight_data and pilot table and lists the pilots that flew

while in non-qualified status

*/

SELECT flight_data. Flight_id, pilot.pilot_id, pilot.qualified

FROM flight_data

INNER JOIN pilot ON pilot.pilot_id = flight_data.pilot_id

WHERE pilot.qualified = '0';

/* 2. Update multiple pilot's qualification using WHERE clause

The query also demonstrates the capability of updating multiple qualification status

at once

*/

SELECT * FROM pilot

WHERE pilot.qualified = 0;

UPDATE pilot

SET qualified = '1'

WHERE qualified = '0';

SELECT * FROM pilot

WHERE pilot.qualified = 0;

ROLLBACK;

/* 3. Count flights flown by pilots assigned to a particular squadron

The query joins 3 tables using flight_data, squadron, and pilot tables to list flights that were flown by pilots from a specific squadron_id

*/

```
SELECT COUNT(flight_data.flight_id)
FROM flight_data
INNER JOIN pilot ON pilot.pilot_id = flight_data.pilot_id
INNER JOIN squadron ON squadron.squadron_id = pilot.squadron_id
WHERE squadron.squadron_id = '100495';
```

/* 4. Select aircraft that had flown 2 or more flights

The query demonstrates a combination of JOIN, GROUP BY, and HAVING function to list aircraft that has flown at least two flights. The query displays the tail_number and number of flights per aircraft

*/

```
SELECT aircraft.tail_number, COUNT(flight_data.flight_id) AS FlightsFlown
FROM aircraft
INNER JOIN flight_data ON flight_data.aircraft_id = aircraft.aircraft_id
GROUP BY aircraft.tail_number
HAVING COUNT(flight_data.flight_id) >= 2;
```

/* 5. Select squadron with the highest number of pilots

Another example of JOIN, GROUP BY, and HAVING functions used in a query to list the squadron with the most number of pilots

*/

```
SELECT squadron.squadron_name, COUNT(pilot.pilot_id) AS Pilots
FROM squadron
INNER JOIN pilot ON pilot.squadron_id = squadron.squadron_id
GROUP BY squadron.squadron_name
HAVING COUNT(pilot.pilot_id) > 3;
```

/* 6. Count the flights flown by each pilot and sort from highest to lowest

This query demonstrates GROUP BY with SORT function listing the flights by pilots in descending order

*/

```
SELECT pilot_id, COUNT(flight_id) AS FlightsFlown
FROM flight_data
GROUP BY pilot_id
ORDER BY FlightsFlown DESC;
```

/* 7. List all squadrons where at least 3 pilots are assigned that have flown a sortie

This query demonstrates multiple aggregation with 3 table JOINS and using GROUP BY and HAVING functions to list squadrons with at least 3 pilots assigned that have flown a flight

*/

```
SELECT squadron.squadron_name, COUNT (DISTINCT (pilot.pilot_id)) AS Pilots,
       COUNT (DISTINCT(flight_data.flight_id)) as Flights
FROM squadron
INNER JOIN pilot ON squadron.squadron_id = pilot.squadron_id
INNER JOIN flight_data ON flight_data.pilot_id = pilot.pilot_id
GROUP BY squadron.squadron_name
HAVING COUNT(pilot.pilot_id) >= 3;
```

/* 8. List the pilots name that have flown at night on a specific day

This query demonstrates a JOIN with multi-variable WHERE clause to display pilots that have flown night sorties

*/

```
SELECT pilot.first_name, pilot.last_name, flight_data.flight_julian_date, flight_data.takeoff_time
FROM pilot
INNER JOIN flight_data ON flight_data.pilot_id = pilot.pilot_id
```

```
WHERE flight_data.flight_julian_date = '22339' AND  
      flight_data.takeoff_time >= '17:00';
```

DDL, DML, and Query Output

Table MAINTENANCE dropped.

Table AIRCRAFT_STATUS dropped.

Table FLIGHT_DATA dropped.

Table PILOT dropped.

Table AIRCRAFT dropped.

Table SQUADRON dropped.

View AIRCRAFTCONFIG dropped.

View PILOTRANK dropped.

View LANDINGSTATUS dropped.

Sequence SEQ_AIRCRAFT_AIRCRAFT_ID dropped.

Sequence SEQ_MAINTENANCE_JOB_ID dropped.

Sequence SEQ_PILOT_PILOT_ID dropped.

Table SQUADRON created.

Table AIRCRAFT created.

Table PILOT created.

Table FLIGHT_DATA created.

Table AIRCRAFT_STATUS created.

Table MAINTENANCE created.

Table AIRCRAFT altered.

Table PILOT altered.

Table SQUADRON altered.

Table MAINTENANCE altered.

Table AIRCRAFT_STATUS altered.

Table FLIGHT_DATA altered.

Index IDX_TAIL created.

Index IDX_WING created.

Index IDX_FLIGHT created.

Index IDX_MAINTENANCE created.

Index IDX_STATUS created.

Index IDX_PILOT created.

Index IDX_SQUADRON created.

View AIRCRAFTCONFIG created.

View PILOTRANK created.

View LANDINGSTATUS created.

Sequence SEQ_AIRCRAFT_AIRCRAFT_ID created.

Sequence SEQ_MAINTENANCE_JOB_ID created.

Sequence SEQ_PILOT_PILOT_ID created.

Trigger TRG_FLIGHT compiled

Trigger TRG_MAINTENANCE compiled

Trigger TRG_STATUS compiled

Trigger TRG_AIRCRAFT compiled

Trigger TRG_PILOT compiled

Trigger TRG_SQUADRON compiled

TABLE_NAME

COURSE

ENROLLMENT

GRADE

GRADE_CONVERSION

GRADE_TYPE

GRADE_TYPE_WEIGHT

INSTRUCTOR

SECTION

STUDENT

ZIPCODE

SQUADRON

TABLE_NAME

AIRCRAFT

PILOT

FLIGHT_DATA

AIRCRAFT_STATUS

MAINTENANCE

16 rows selected.

OBJECT_NAME

STATUS CREATED

LAST_DDL_

-----INSTRUCTOR_ID_SEQ
16-FEB-22

VALID 16-FEB-22

SECTION_ID_SEQ
FEB-22

VALID 16-FEB-22 16-

STUDENT_ID_SEQ
FEB-22

VALID 16-FEB-22 16-

COURSE_NO_SEQ
FEB-22

VALID 16-FEB-22 16-

COURSE
22

VALID 16-FEB-22 16-FEB-

CRSE_CRSE_FK_I
FEB-22

VALID 16-FEB-22 16-

CRSE_PK
22

VALID 16-FEB-22 16-FEB-

ENROLLMENT
FEB-22

VALID 16-FEB-22 16-

ENR_SECT_FK_I FEB-22	VALID 16-FEB-22 16-
ENR_PK 22	VALID 16-FEB-22 16-FEB-
GRADE	VALID 16-FEB-22 16-FEB-22
OBJECT_NAME LAST_DDL_	STATUS CREATED

GR_GRTW_FK_I FEB-22	VALID 16-FEB-22 16-
GR_PK	VALID 16-FEB-22 16-FEB-22
GRADE_CONVERSION 16-FEB-22	VALID 16-FEB-22
GRCON_PK 22	VALID 16-FEB-22 16-FEB-
GRADE_TYPE FEB-22	VALID 16-FEB-22 16-
GRTYP_PK 22	VALID 16-FEB-22 16-FEB-
GRADE_TYPE_WEIGHT 16-FEB-22	VALID 16-FEB-22
GRTW_GRTYP_FK_I FEB-22	VALID 16-FEB-22 16-
GRTW_PK 22	VALID 16-FEB-22 16-FEB-
INSTRUCTOR FEB-22	VALID 16-FEB-22 16-
INST_ZIP_FK_I FEB-22	VALID 16-FEB-22 16-
OBJECT_NAME LAST_DDL_	STATUS CREATED

INST_PK 22	VALID 16-FEB-22 16-FEB-
SECTION 22	VALID 16-FEB-22 16-FEB-
SECT_CRSE_FK_I FEB-22	VALID 16-FEB-22 16-
SECT_INST_FK_I FEB-22	VALID 16-FEB-22 16-
SECT_PK 22	VALID 16-FEB-22 16-FEB-
SECT_SECT2_UK FEB-22	VALID 16-FEB-22 16-
STUDENT 22	VALID 16-FEB-22 16-FEB-
STU_ZIP_FK_I FEB-22	VALID 16-FEB-22 16-
STU_PK	VALID 16-FEB-22 16-FEB-22
ZIPCODE 22	VALID 16-FEB-22 16-FEB-
ZIP_PK	VALID 16-FEB-22 16-FEB-22
OBJECT_NAME LAST_DDL_	STATUS CREATED

AIRCRAFTCONFIG DEC-22	VALID 10-DEC-22 10-
PILOTTRANK DEC-22	VALID 10-DEC-22 10-
LANDINGSTATUS DEC-22	VALID 10-DEC-22 10-
TRG_FLIGHT DEC-22	VALID 10-DEC-22 10-
TRG_MAINTENANCE 10-DEC-22	VALID 10-DEC-22

TRG_PILOT 22	VALID 10-DEC-22 10-DEC-
TRG_STATUS DEC-22	VALID 10-DEC-22 10-
TRG_AIRCRAFT DEC-22	VALID 10-DEC-22 10-
TRG_SQUADRON DEC-22	VALID 10-DEC-22 10-
SQUADRON DEC-22	VALID 10-DEC-22 10-
PK_SQUADRON DEC-22	VALID 10-DEC-22 10-
OBJECT_NAME LAST_DDL_	STATUS CREATED

AIRCRAFT 22	VALID 10-DEC-22 10-DEC-
PK_AIRCRAFT DEC-22	VALID 10-DEC-22 10-
PILOT	VALID 10-DEC-22 10-DEC-22
PK_PILOT 22	VALID 10-DEC-22 10-DEC-
FLIGHT_DATA DEC-22	VALID 10-DEC-22 10-
PK_FLIGHT_DATA DEC-22	VALID 10-DEC-22 10-
AIRCRAFT_STATUS DEC-22	VALID 10-DEC-22 10-
MAINTENANCE DEC-22	VALID 10-DEC-22 10-
PK_MAINTENANCE 10-DEC-22	VALID 10-DEC-22
IDX_TAIL 22	VALID 10-DEC-22 10-DEC-

IDX_WING
22

VALID 10-DEC-22 10-DEC-

OBJECT_NAME
LAST_DDL_

STATUS CREATED

IDX_FLIGHT
22

VALID 10-DEC-22 10-DEC-

IDX_MAINTENANCE
10-DEC-22

VALID 10-DEC-22

IDX_STATUS
DEC-22

VALID 10-DEC-22 10-

IDX_PILOT
22

VALID 10-DEC-22 10-DEC-

IDX_SQUADRON
DEC-22

VALID 10-DEC-22 10-

SEQ_AIRCRAFT_AIRCRAFT_ID
22 10-DEC-22

VALID 10-DEC-

SEQ_MAINTENANCE_JOB_ID
22 10-DEC-22

VALID 10-DEC-

SEQ_PILOT_PILOT_ID
10-DEC-22

VALID 10-DEC-22

63 rows selected.

Commit complete.

1 row inserted.

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1 row inserted.

Commit complete.

SQL Query Script Output

AIRCRAFT_ID MODIFIED_BY	SQUADRON_ID DATE_MODI	CONFI	GEOLOC	MISSION_DE TAIL_	CREATED_BY	DATE_CREA
34205589 10-DEC-22	100495 XX001	MSET	F035A	A5589	DBST_USER	10-DEC-22 DBST_USER
34205583 10-DEC-22	100495 XX001	MSET	F035A	A5583	DBST_USER	10-DEC-22 DBST_USER
34205581 10-DEC-22	100496 XX001	MSET	F035A	A5581	DBST_USER	10-DEC-22 DBST_USER
34205571 10-DEC-22	100496 XX001	MSET	F035A	A5571	DBST_USER	10-DEC-22 DBST_USER
34205572 10-DEC-22	100495 XX001	MSET	F035A	A5572	DBST_USER	10-DEC-22 DBST_USER
34205570 10-DEC-22	100496 XX001	MSET	F035A	A5570	DBST_USER	10-DEC-22 DBST_USER

34195486 10-DEC-22	100496 XX001 MSET	F035A	A5486 DBST_USER	10-DEC-22 DBST_USER
34195485 10-DEC-22	100495 XX001 MSET	F035A	A5485 DBST_USER	10-DEC-22 DBST_USER
34195484 10-DEC-22	100496 XX001 MSET	F035A	A5484 DBST_USER	10-DEC-22 DBST_USER
34195475 10-DEC-22	100497 XX001 MSET	F035A	A5475 DBST_USER	10-DEC-22 DBST_USER
34195476 10-DEC-22	100497 XX001 MSET	F035A	A5476 DBST_USER	10-DEC-22 DBST_USER

AIRCRAFT_ID MODIFIED_BY	SQUADRON_ID DATE_MODI	CONFI	GEOLOC	MISSION_DE TAIL_	CREATED_BY	DATE_CREA
----------------------------	--------------------------	-------	--------	---------------------	------------	-----------

34195474 10-DEC-22	100497 XX001 MSET	F035A	A5474 DBST_USER	10-DEC-22 DBST_USER
34195473 10-DEC-22	100497 XX001 MSET	F035A	A5473 DBST_USER	10-DEC-22 DBST_USER
34195495 10-DEC-22	100495 XX001 MSET	F035A	A5495 DBST_USER	10-DEC-22 DBST_USER
34205580 10-DEC-22	100496 XX001 MSET	F035A	A5580 DBST_USER	10-DEC-22 DBST_USER

15 rows selected.

RAN	FIRST_NAME	LAST_NAME	PILOT_I Q
-----	------------	-----------	-----------

O03	Richard	Galero	1234567 1
O02	James	Rivas	0000002 1
O04	Kyle	Ward	0000003 1
O05	Joseph	Miles	0000004 1
O05	Joel	Banks	0000005 1
O03	Benjamin	Lovett	0000006 1

O04 Paul	Walker	0000007 0
O03 Leila	Thompson	0000008 1
O03 Jonathan	Dayao	0000009 1
O05 Roseanne	Garcia	0000010 1
O03 Kristin	Lopez	0000011 0

11 rows selected.

LAST_NAME	FIRST_NAME	RAN Q
-----	-----	-----
Galero	Richard	O03 1
Rivas	James	O02 1
Ward	Kyle	O04 1
Miles	Joseph	O05 1
Banks	Joel	O05 1
Lovett	Benjamin	O03 1
Thompson	Leila	O03 1
Dayao	Jonathan	O03 1
Garcia	Roseanne	O05 1

9 rows selected.

FLIGHT_ID	AIRCRAFT_ID	PILOT_I	FLIGHT_JULIAN_DATE	FLIGHT_NUMBER	LAND	LANDING_STATUS
LANDI	TAKE	TAKEO	CREATED_BY	DATE_CREA	MODIFIED_BY	DATE_MODI
AIRCRAFT_ID	SQUADRON_ID	CONFI	GEOLOC	MISSION_DE	TAIL_CREATED_BY	DATE_CREA
MODIFIED_BY	DATE_MODI					

22339801	34195475	0000002	22339	801 EGUL	1 14:32 EGUL 13:00 DBST_USER
10-DEC-22 DBST_USER			10-DEC-22	34195475	100497 XX001 MSET F035A A5475
DBST_USER	10-DEC-22 DBST_USER			10-DEC-22	
22339802	34195486	0000003	22339	802 EGUL	1 14:33 EGUL 13:00 DBST_USER
10-DEC-22 DBST_USER			10-DEC-22	34195486	100496 XX001 MSET F035A A5486
DBST_USER	10-DEC-22 DBST_USER			10-DEC-22	
22339803	34205572	0000004	22339	803 EGUL	1 14:32 EGUL 13:00 DBST_USER
10-DEC-22 DBST_USER			10-DEC-22	34205572	100495 XX001 MSET F035A A5572
DBST_USER	10-DEC-22 DBST_USER			10-DEC-22	
22339804	34195474	0000005	22339	804 EGUL	1 14:27 EGUL 13:00 DBST_USER
10-DEC-22 DBST_USER			10-DEC-22	34195474	100497 XX001 MSET F035A A5474
DBST_USER	10-DEC-22 DBST_USER			10-DEC-22	
22339805	34195476	0000006	22339	805 EGUL	1 14:50 EGUL 13:00 DBST_USER
10-DEC-22 DBST_USER			10-DEC-22	34195476	100497 XX001 MSET F035A A5476
DBST_USER	10-DEC-22 DBST_USER			10-DEC-22	
22339806	34205581	0000007	22339	806 EGUL	1 14:42 EGUL 13:06 DBST_USER
10-DEC-22 DBST_USER			10-DEC-22	34205581	100496 XX001 MSET F035A A5581
DBST_USER	10-DEC-22 DBST_USER			10-DEC-22	
22339807	34195495	0000008	22339	807 EGUL	1 14:41 EGUL 13:04 DBST_USER
10-DEC-22 DBST_USER			10-DEC-22	34195495	100495 XX001 MSET F035A A5495
DBST_USER	10-DEC-22 DBST_USER			10-DEC-22	
22339808	34205571	0000009	22339	808 EGUL	1 14:41 EGUL 13:05 DBST_USER
10-DEC-22 DBST_USER			10-DEC-22	34205571	100496 XX001 MSET F035A A5571
DBST_USER	10-DEC-22 DBST_USER			10-DEC-22	
22339809	34205580	0000010	22339	809 EGUL	1 14:42 EGUL 13:08 DBST_USER
10-DEC-22 DBST_USER			10-DEC-22	34205580	100496 XX001 MSET F035A A5580
DBST_USER	10-DEC-22 DBST_USER			10-DEC-22	
22339810	34205589	0000011	22339	810 EGUL	1 14:42 EGUL 13:08 DBST_USER
10-DEC-22 DBST_USER			10-DEC-22	34205589	100495 XX001 MSET F035A A5589
DBST_USER	10-DEC-22 DBST_USER			10-DEC-22	
22339811	34195475	0000002	22339	811 EGUL	1 17:20 EGUL 19:10 DBST_USER
10-DEC-22 DBST_USER			10-DEC-22	34195475	100497 XX001 MSET F035A A5475
DBST_USER	10-DEC-22 DBST_USER			10-DEC-22	

FLIGHT_ID	AIRCRAFT_ID	PILOT_I	FLIGHT_JULIAN_DATE	FLIGHT_NUMBER	LAND	LANDING_STATUS
LANDI	TAKE	TAKEO	CREATED_BY	DATE_CREA	MODIFIED_BY	DATE_MODI
AIRCRAFT_ID	SQUADRON_ID	CONFI	GEOLOC	MISSION_DE	TAIL_CREATED_BY	DATE_CREA
MODIFIED_BY		DATE_MODI				

```

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-----
-----
22339812 34195486 0000003      22339      812 EGUL      1 17:20 EGUL 19:10 DBST_USER
10-DEC-22 DBST_USER      10-DEC-22 34195486      100496 XX001 MSET      F035A      A5486
DBST_USER      10-DEC-22 DBST_USER      10-DEC-22

22339813 34195474 0000004      22339      813 EGUL      1 17:20 EGUL 19:10 DBST_USER
10-DEC-22 DBST_USER      10-DEC-22 34195474      100497 XX001 MSET      F035A      A5474
DBST_USER      10-DEC-22 DBST_USER      10-DEC-22

22339814 34195476 0000005      22339      814 TYFR      1 17:20 EGUL 20:10 DBST_USER
10-DEC-22 DBST_USER      10-DEC-22 34195476      100497 XX001 MSET      F035A      A5476
DBST_USER      10-DEC-22 DBST_USER      10-DEC-22

22339815 34205572 0000006      22339      815 EGUL      1 17:34 EGUL 19:23 DBST_USER
10-DEC-22 DBST_USER      10-DEC-22 34205572      100495 XX001 MSET      F035A      A5572
DBST_USER      10-DEC-22 DBST_USER      10-DEC-22

22339816 34205581 0000007      22339      816 EGUL      1 17:34 EGUL 19:23 DBST_USER
10-DEC-22 DBST_USER      10-DEC-22 34205581      100496 XX001 MSET      F035A      A5581
DBST_USER      10-DEC-22 DBST_USER      10-DEC-22

22339817 34205580 0000008      22339      817 EGUL      1 17:30 EGUL 19:17 DBST_USER
10-DEC-22 DBST_USER      10-DEC-22 34205580      100496 XX001 MSET      F035A      A5580
DBST_USER      10-DEC-22 DBST_USER      10-DEC-22

22339818 34205571 0000009      22339      818 EGUL      1 17:30 EGUL 19:25 DBST_USER
10-DEC-22 DBST_USER      10-DEC-22 34205571      100496 XX001 MSET      F035A      A5571
DBST_USER      10-DEC-22 DBST_USER      10-DEC-22

```

18 rows selected.

```

SQUADRON_ID COUNTRY      MAJOR_COMMAND      SQUADRON_NAME
STATE      WING      CREATED_BY      DATE_CREA MODIFIED_BY
DATE_MODI

```

```

-----
-----
-----
100001 USA      ACC      First Fighter Squadron      VA
First Fighter Wing      DBST_USER      10-DEC-22 DBST_USER      10-DEC-
22

```


100004 USA	ACC	4th Fighter Squadron
North Carolina	4th Fighter Wing	DBST_USER
10-DEC-22		10-DEC-22 DBST_USER
100052 Germany	USAFE	52nd Fighter Squadron
N/A	52nd Fighter Wing	DBST_USER
10-DEC-22		10-DEC-22 DBST_USER
100086 Germany	USAFE	786th Airlift Squadron
N/A	86th Airlift Wing	DBST_USER
10-DEC-22		10-DEC-22 DBST_USER
100388 USA	ACC	388th Fighter Squadron
Utah	388th Fighter Wing	DBST_USER
10-DEC-22		10-DEC-22 DBST_USER
100419 USA	ACC	419th Fighter Squadron
N/A	419th Fighter Wing	DBST_USER
10-DEC-22		10-DEC-22 DBST_USER
100495 UK	USAFE	495th Fighter Squadron
N/A	48th Fighter Wing	DBST_USER
10-DEC-22		10-DEC-22 DBST_USER
100496 UK	USAFE	496th Fighter Squadron
N/A	48th Fighter Wing	DBST_USER
10-DEC-22		10-DEC-22 DBST_USER
100497 UK	USAFE	497th Fighter Squadron
N/A	48th Fighter Wing	DBST_USER
10-DEC-22		10-DEC-22 DBST_USER
100498 UK	USAFE	498th Fighter Squadron
N/A	48th Fighter Wing	DBST_USER
10-DEC-22		10-DEC-22 DBST_USER

10 rows selected.

TAIL_STATU AIRCRAFT_S DISCREPANCY_NARRATIVE
JOB_NUMBER

A5583 NMC 42000 VMS Flight control inop
223391201

A5583 NMC	42001	aircraft anels require removal
223391202		
A5583 NMC	24000	Flight data recorder was intermittent
223391203		
A5570 PMC	04000	400 Hour inspection due
223391207		
A5570 PMC	04100	400 Hour inspection due
223391208		
A5570 PMC	04500	Inspect engine bay
223391209		
A5570 PMC	04800	test hydraulic flaps
223391210		
A5570 PMC	04900	aircraft wash
223391211		
A5486 NMC	13000	landing gear requires inspection
223391204		
A5486 NMC	13100	landing gear requires inspection
223391205		
A5486 NMC	13110	landing gear requires inspection
223391206		

11 rows selected.

TAIL_ AIRCRAFT_ID STATU

A5583	34205583	NMC
A5570	34205570	PMC
A5486	34195486	NMC

SQUADRON_ID AIRCRAFTBYSQ

100496	6
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100495	5
100497	4

SQUADRON_ID	SQUADRON_NAME	COUNTRY
100495	495th Fighter Squadron	UK
100496	496th Fighter Squadron	UK
100497	497th Fighter Squadron	UK
100498	498th Fighter Squadron	UK
100086	786th Airlift Squadron	Germany
100052	52nd Fighter Squadron	Germany

6 rows selected.

LENGTH(SQUADRON_NAME)

22
22
22
22
22
22
20
21
22
22

10 rows selected.

FLIGHT_ID	AIRCRAFT_ID	PILOT_I	FLIGHT_JULIAN_DATE	FLIGHT_NUMBER	LAND	LANDING_STATUS
LANDI	TAKE	TAKEO	CREATED_BY	DATE_CREA	MODIFIED_BY	DATE_MODI

22339801	34195475	00000002	22339	801 EGUL	1 14:32 EGUL	13:00 DBST_USER
10-DEC-22	DBST_USER		10-DEC-22			

1 row deleted.

no rows selected

Rollback complete.

FLIGHT_ID	AIRCRAFT_ID	PILOT_I	FLIGHT_JULIAN_DATE	FLIGHT_NUMBER	LAND	LANDING_STATUS
LANDI	TAKE	TAKEO	CREATED_BY	DATE_CREA	MODIFIED_BY	DATE_MODI

22339801	34195475	00000002	22339	801 EGUL	1 14:32 EGUL	13:00 DBST_USER
10-DEC-22	DBST_USER		10-DEC-22			

SQUADRON_ID	COUNTRY	MAJOR_COMMAND	SQUADRON_NAME
STATE	WING	CREATED_BY	DATE_CREA
DATE_MODI		MODIFIED_BY	

100086	Germany	USAFE	786th Airlift Squadron
N/A	86th Airlift Wing	DBST_USER	10-DEC-22 DBST_USER
10-DEC-22			

100052	Germany	USAFE	52nd Fighter Squadron
N/A	52nd Fighter Wing	DBST_USER	10-DEC-22 DBST_USER
10-DEC-22			

1 row updated.

SQUADRON_ID	COUNTRY	MAJOR_COMMAND	SQUADRON_NAME
STATE	WING	CREATED_BY	DATE_CREA MODIFIED_BY
DATE_MODI			

100086	Germany	USAFE	786th Airlift Squadron
N/A	86th Airlift Wing	DBST_USER	10-DEC-22 DBST_USER
10-DEC-22			

100052	Germany	ACC	52nd Fighter Squadron
N/A	52nd Fighter Wing	DBST_USER	10-DEC-22 DBST_USER
10-DEC-22			

Rollback complete.

SQUADRON_ID	COUNTRY	MAJOR_COMMAND	SQUADRON_NAME
STATE	WING	CREATED_BY	DATE_CREA MODIFIED_BY
DATE_MODI			

100086	Germany	USAFE	786th Airlift Squadron
N/A	86th Airlift Wing	DBST_USER	10-DEC-22 DBST_USER
10-DEC-22			

100052	Germany	USAFE	52nd Fighter Squadron
N/A	52nd Fighter Wing	DBST_USER	10-DEC-22 DBST_USER
10-DEC-22			

FLIGHT_ID PILOT_I Q

22339806 0000007 0

22339810 0000011 0

22339816 0000007 0

FIRST_NAME	LAST_NAME	PILOT_I	SQUADRON_ID	Q	RAN	CREATED_BY
DATE_CREA	MODIFIED_BY	DATE_MODI				

Paul	Walker	0000007	100495 0 004	DBST_USER	10-DEC-22
DBST_USER	10-DEC-22				
Kristin	Lopez	0000011	100496 0 003	DBST_USER	10-DEC-22
DBST_USER	10-DEC-22				

2 rows updated.

no rows selected

Rollback complete.

COUNT(FLIGHT_DATA.FLIGHT_ID)

5

TAIL_ FLIGHTSFLOWN

A5474	2
A5475	2
A5580	2
A5572	2
A5476	2
A5571	2
A5486	2
A5581	2

8 rows selected.

SQUADRON_NAME	PILOTS
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495th Fighter Squadron	4
496th Fighter Squadron	4

PILOT_I FLIGHTSFLOWN

0000003	2
0000008	2
0000009	2
0000002	2
0000006	2
0000007	2
0000004	2
0000005	2
0000011	1

0000010 1

10 rows selected.

SQUADRON_NAME	PILOTS	FLIGHTS
495th Fighter Squadron	3	5
496th Fighter Squadron	4	7
497th Fighter Squadron	3	6

FIRST_NAME	LAST_NAME	FLIGHT_JULIAN_DATE TAKEO
James	Rivas	22339 19:10
Kyle	Ward	22339 19:10
Joseph	Miles	22339 19:10
Joel	Banks	22339 20:10
Benjamin	Lovett	22339 19:23
Paul	Walker	22339 19:23
Leila	Thompson	22339 19:17
Jonathan	Dayao	22339 19:25

8 rows selected.

DATABASE ADMINISTRATION AND MONITORING

Roles and Responsibilities

- **Database Administrator** will be the primary responsible for access and maintenance of the database. They will be the main point of contact to give access to users and perform routine maintenance of the system. This role will be fulfilled by the Maintenance Management Analysis section within the organization.
- **System Administrator** role will be levied upon the Major Command Analyst located in the USAFE Headquarters. This role will perform functions supporting server maintenance, operating software, and any support that involves collaboration with the Air Force Communication Squadrons.
- **Security Administrator** will monitor login records to ensure the system is not breached and potential leak of information is prevented. They will perform security and vulnerability checks as deemed necessary or as dictated by the Air Force security standards.

System Information

DBMS: Oracle Database Version 19.1.0 // 12.2.0.3

System Requirements:

- Minimum 2 CPU Cores per server
- Minimum 32 GB RAM per server
- Minimum 1 TB hard disk space per server

Performance Monitoring and Database Efficiency

Performance monitoring shall be performed jointly by the database administrator and the system administrator. Use of automated tools to track the performance and efficiency of the database is highly

encouraged. Furthermore, daily checks and balances to ensure that the database is performing up to specification will be performed.

Data Formats

The database utilizes different data formats such as *strings, integers, date and time* formats. Data exports will be kept at a comma separated value (CSV) format to minimize the impact of time and processing resources required. Application connections to external dashboards will also utilize and push .csv files and shall be processed and transformed external to the database.

Backup and Recovery

The database will be saved and backup twice daily every 06:00 and 18:00. This will ensure that impact to the users is kept at a minimum. This function will be performed and monitored by the system administrator and notify pertinent database managers of any issues. Complete backup of the database will be performed every Sunday at 12:00 P.M. In case of major loss of data, the database will be rolled back to the most recent weekly backup.

References

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